120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 11: Portfolio Optimization and Risk Management using ML

Day 71: Introduction to Portfolio Theory

1 Introduction

Portfolio theory studies how investors can allocate capital across multiple assets to balance return and risk.

Instead of evaluating assets individually, portfolio theory focuses on the collective behavior of assets within a portfolio.

The key idea is diversification: combining assets with imperfect correlations can reduce overall portfolio risk.

This framework was formalized by Harry Markowitz in 1952 and is known as **Modern Portfolio Theory (MPT)**.

2 Portfolio Weights

Assume we invest in n assets.

Let the portfolio weights be:

$$w = (w_1, w_2, ..., w_n)$$

where

- w_i represents the fraction of capital invested in asset i
- $\sum_{i=1}^{n} w_i = 1$

3 Expected Portfolio Return

Let

$$\mu = (\mu_1, \mu_2, ..., \mu_n)$$

be the expected returns of the assets.

The expected portfolio return is:

$$E[R_p] = \sum_{i=1}^{n} w_i \mu_i$$

In vector form:

$$E[R_p] = w^T \mu$$

4 Portfolio Risk

Risk is measured using the variance of portfolio returns.

Let Σ denote the covariance matrix of asset returns:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix}$$

The portfolio variance is:

$$Var(R_p) = w^T \Sigma w$$

The portfolio standard deviation is:

$$\sigma_p = \sqrt{w^T \Sigma w}$$

5 Diversification

Diversification reduces risk when asset returns are not perfectly correlated.

For a two-asset portfolio:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12}$$

where ρ_{12} is the correlation between the assets.

If correlation is low, portfolio risk decreases.

6 Mean-Variance Optimization

The central problem of portfolio theory is:

$$\min_w w^T\Sigma w$$

subject to

$$w^T\mu = R_{target}$$

$$\sum w_i = 1$$

This finds the portfolio with the minimum risk for a given expected return.

7 Efficient Frontier

Solving the optimization problem for different target returns produces the **efficient frontier**.

The efficient frontier represents the set of optimal portfolios that offer the highest expected return for each level of risk.

Portfolios below the frontier are suboptimal because higher returns could be achieved for the same level of risk.

8 Sharpe Ratio

The Sharpe Ratio measures risk-adjusted return:

$$Sharpe = \frac{E[R_p] - R_f}{\sigma_p}$$

where

- R_f is the risk-free rate
- σ_p is portfolio volatility

Maximizing the Sharpe Ratio identifies the **tangency portfolio**.

9 Conclusion

Portfolio theory provides a mathematical framework for diversification and risk management.

The key insights include:

- Portfolio risk depends on correlations between assets
- Diversification can reduce overall risk
- Optimal portfolios lie on the efficient frontier

However, traditional portfolio theory relies on estimates of expected returns and covariances, which can be unstable in financial markets.

These limitations motivate the use of machine learning techniques for portfolio optimization.